MACHINE LEARNING FOR HEAVY METAL DETECTION

Research-paper analysis, model comparison, and a reproducible validation framework

Prepared for: GitHub project documentation

Scope and accuracy note

This document synthesizes published research on ML methods used to estimate or classify heavy-metal contamination, emphasizing soil/environmental datasets. Your exact medium (water, soil, food, sensor or spectroscopy), dataset, target metals and model outputs were not supplied. This PDF therefore does not claim that models have already been trained or validated on your project data, and contains no invented project scores.

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1. Executive summary

Machine learning can support heavy-metal monitoring by learning relationships between measured inputs—such as soil properties, environmental covariates, spectral reflectance or sensor readings—and reference laboratory concentrations or contamination labels. It should be treated as an estimation/screening layer, not a replacement for validated analytical chemistry.

No single algorithm is universally best. Random Forest (RF), Support Vector Regression (SVR/SVM), gradient boosting (including XGBoost), Extra Trees, Partial Least Squares Regression (PLSR), and neural networks all appear in the literature. Performance depends on the target metal, sample size, input modality, concentration range, sampling design and validation protocol.

A 2026 review of 91 industrial-site soil studies reported RF as the best model in 54.29% of studies using environmental covariates. For hyperspectral inputs, RF, SVM and PLSR were among the most frequently used methods. This is a literature trend, not a guarantee RF will win on your dataset. [1]

Recommended benchmark: include a simple baseline, then compare RF, Extra Trees, SVR, Gradient Boosting/XGBoost, and PLSR if inputs are spectral. Use regression for continuous concentration and classification only for a clearly defined contamination label.

“Detection” may mean concentration estimation, contaminated/not-contaminated classification, or identifying which metal is present. Define the task before choosing metrics.

2. Define the task

- **Medium:** soil, groundwater/surface water, food, biological sample, or another matrix.
- **Reference label:** laboratory concentration and method (e.g. ICP-MS/AAS), sensor reading, or expert label.
- **Targets:** Pb, Cd, As, Cr, Hg, Ni, Cu, Zn, etc.; record units such as mg/kg or µg/L.
- **Inputs:** pH, conductivity, organic matter, spectral bands, coordinates, sensor signals, or other variables.
- **Output:** regression for concentration; classification for a predefined contamination/risk category.
- **Sampling structure:** record site, date, batch, instrument and repeated measurements.

Do not use target-derived information, post-test laboratory results, or features unavailable at deployment. This is data leakage and can inflate apparent accuracy.

3. Model comparison

Model	Strengths	Limitations	Role in benchmark
Dummy baseline	Shows whether ML adds value.	Does not learn feature relationships.	Required baseline.
Linear / Ridge	Fast, interpretable baseline.	May miss nonlinear patterns; scaling often helps.	Regression sanity check.
PLSR	Useful for correlated, high-dimensional spectral data.	Tune components inside CV; may miss nonlinearities.	Prioritize for spectroscopy.
Random Forest	Learns nonlinear patterns/interactions; strong tabular baseline.	May smooth extremes; feature importance can mislead.	Good default for environmental tables.
Extra Trees	Randomized tree ensemble; often competitive.	May underfit some patterns; tune parameters.	Compare with RF.
SVR (RBF)	Flexible nonlinear model for small/medium datasets.	Needs scaling; tune C, epsilon and gamma.	Nonlinear benchmark.
Gradient Boosting / XGBoost	Can model complex nonlinear relationships.	Tuning needed; may overfit small/noisy data.	Candidate, not assumed winner.
MLP / neural network	Flexible with sufficient data.	Needs scaling/tuning; may be unstable on small data.	Optional for larger datasets.
Classifier	Predicts a predefined class/risk category.	Depends on threshold and class imbalance.	Only for categorical targets.

For a fair comparison, use the same outer train/test split for every candidate, fit preprocessing only on training folds, tune only on training data, and evaluate every model on the same held-out samples.

4. Research-paper analysis

[1] Monitoring heavy metal(loid) concentrations in soils of industrially contaminated sites using machine learning models. *Journal of Hazardous Materials*, 502 (2026), 141011. DOI: 10.1016/j.jhazmat.2025.141011. Review of 91 studies (2014–2025), separating environmental covariates from hyperspectral inputs. RF was most often optimal for environmental-covariate modeling; RF, SVM and PLSR were frequent for hyperspectral data. Implication: choose candidates based on input modality. PubMed record.

[2] Predicting Heavy Metal Adsorption on Soil with Machine Learning and Mapping Global Distribution of Soil Adsorption Capacities. *Environmental Science & Technology* (2021). DOI: 10.1021/acs.est.1c02479. Compiled 4,420 data points from 1,105 soils across 150 papers for adsorption of Cd, Cr, Cu, Pb, Ni and Zn. Gradient-boosting decision trees performed best for the combined adsorption model; metal-specific models were also developed and SHAP used for feature attribution. Caution: adsorption capacity is not identical to direct field concentration detection. Publisher page.

[3] A general methodological framework for predicting and assessing heavy metal pollution in paddy soils using machine learning models. (2025), open access: PMC full text. Compared RF, Extra Trees, XGBoost and Gradient Boosting across multiple metals. RF performed well for several elements while Extra Trees performed well for others, supporting per-metal evaluation rather than declaring one universal winner.

[4] Estimation of Heavy Metal Content in Soil Based on Machine Learning Models. *Land*, 11(7), 1037 (2022). DOI: 10.3390/land11071037. Compared a LASSO–genetic algorithm–back-propagation neural network with SVR, RF and spatial interpolation using remote-sensing and field variables. Hybrid complexity should be justified by reproducible gains over simpler baselines. Full text.

[5] Comparative analysis of different machine learning algorithms for predicting trace metal concentrations in soils under intensive paddy cultivation. *Computers and Electronics in Agriculture*, 219 (2024), 108772. DOI: 10.1016/j.compag.2024.108772. Examined several ML methods and input variables.

Cross-paper conclusion: RF, SVR/SVM, boosting and—where appropriate—PLSR or neural networks are reasonable candidates. Scores from different papers are not a controlled head-to-head comparison because datasets, targets, units, inputs and validation differ. Do not copy published scores as project results.

5. Validation and verification protocol

1. Data audit: record data source, unique samples/sites, units, missingness, duplicates, detection limits, target distribution and class balance. Keep repeated measurements of the same physical sample together.

2. Lock a final test set: random split (e.g. 80/20) is appropriate only for independent samples. If observations share sites, dates, batches or spatial proximity, use grouped or spatial splits.

3. Cross-validation: use 5-fold CV as a starting point; use GroupKFold for groups or spatial block CV when predicting new locations. Imputation, scaling, feature selection and dimensionality reduction must occur inside a pipeline and be fitted within each fold.

4. Tune fairly: GridSearchCV or RandomizedSearchCV on training data only. Record parameter search, metric, seed and package versions. Consider nested CV for small datasets with extensive tuning.

5. Final evaluation: fit the chosen pipeline on training data, evaluate once on the untouched test set, and report per-metal results plus sample count. Compare training and test scores for overfitting.

6. Verification: inspect predicted-versus-measured plots, residuals by concentration, errors near decision thresholds and performance across sites/batches/instruments. Verify against laboratory reference measurements before operational use.

6. Metrics and interpretation

Regression: MAE (average absolute error in original units), RMSE (penalizes larger errors) and R^2 (fit relative to test-set mean baseline; may be negative on test data). Report per-metal scores, units and sample counts. Avoid MAPE for zero or near-zero concentrations.

Classification: confusion matrix, precision, recall/sensitivity, specificity, F1, ROC-AUC and PR-AUC when positives are rare. Define any contamination threshold using a cited regulation or documented project protocol before evaluating the test set.

Feature importance / SHAP: describes how a fitted model uses inputs, but does not prove causality. Correlated features can distort importance; discuss uncertainty and limitations.

7. Results table — complete after running your code

No dataset or run logs were supplied, so this template is intentionally blank. Numerical scores have not been invented.

Target	Model	CV MAE	CV RMSE	Test MAE	Test RMSE	Test R^2
Metal/label	Model	—	—	—	—	—
Metal/label	Model	—	—	—	—	—
Metal/label	Model	—	—	—	—	—
Metal/label	Model	—	—	—	—	—
Metal/label	Model	—	—	—	—	—
Metal/label	Model	—	—	—	—	—

8. Limitations and responsible use

Common threats include small/unrepresentative samples, censored values below detection limits, instrument drift, inconsistent laboratory methods, target leakage, spatial leakage and tuning against the test set. Document these explicitly. Laboratory measurements remain the reference standard unless the intended use and applicable regulations establish another accepted method. Do not use an ML model alone to declare a sample safe or unsafe without calibration, external validation and domain review.

9. GitHub checklist and suggested structure

- Cite dataset source and license; include a data dictionary.
- Document medium, target metals, units, reference method and intended use.
- Include split strategy, random seed, package versions and training script.
- Use pipelines to prevent preprocessing leakage.
- Benchmark simple baselines and candidates on identical splits.
- Report per-target CV and test metrics; distinguish internal from external validation.
- Include plots, error analysis, limitations and deployment boundaries.

```
heavy-metal-detection/
  README.md
  docs/research_analysis.pdf
  data/README.md
  notebooks/01_eda.ipynb
  src/train.py
  src/evaluate.py
  results/metrics.csv
  results/figures/
```

Reproducibility statement

This PDF is a literature synthesis and proposed validation plan, not an experimental results paper. Update the results table and conclusions only after training and evaluating models on the project's documented dataset.